Merck
Q2 2024 Earnings

July 30, 2024

Agenda

Strategy and Business Update

Rob Davis
Chairman and Chief Executive Officer

Business/Financial Results and Outlook

Caroline Litchfield
Chief Financial Officer

Research Update

Dr. Dean Li
President, Merck Research Laboratories

Question & Answer Session

Forward-looking statement of Merck & Co., Inc., Rahway, N.J., USA

This presentation of Merck & Co., Inc., Rahway, N.J., USA (the "company") includes "forward-looking statements" within the meaning of the safe harbor provisions of the U.S. Private Securities Litigation Reform Act of 1995. These statements are based upon the current beliefs and expectations of the company's management and are subject to significant risks and uncertainties. There can be no guarantees with respect to pipeline candidates that the candidates will receive the necessary regulatory approvals or that they will prove to be commercially successful. If underlying assumptions prove inaccurate or risks or uncertainties materialize, actual results may differ materially from those set forth in the forward-looking statements.

Risks and uncertainties include but are not limited to, general industry conditions and competition; general economic factors, including interest rate and currency exchange rate fluctuations; the impact of pharmaceutical industry regulation and health care legislation in the United States and internationally; global trends toward health care cost containment; technological advances, new products and patents attained by competitors; challenges inherent in new product development, including obtaining regulatory approval; the company's ability to accurately predict future market conditions; manufacturing difficulties or delays; financial instability of international economies and sovereign risk; dependence on the effectiveness of the company's patents and other protections for innovative products; and the exposure to litigation, including patent litigation, and/or regulatory actions.

The company undertakes no obligation to publicly update any forward-looking statement, whether as a result of new information, future events or otherwise. Additional factors that could cause results to differ materially from those described in the forward-looking statements can be found in the company's Annual Report on Form 10-K for the year ended December 31, 2023 and the company's other filings with the Securities and Exchange Commission (SEC) available at the SEC's Internet site (www.sec.gov).

MERCK
INVENTING FOR LIFE

Strategy and Business Update

Rob Davis
Chairman and Chief Executive Officer

Delivered on our key strategic priorities in Q2 2024

- **Advanced the pipeline to meet patient unmet need**
- **Executed on strategic business development to augment pipeline**
- **Achieved strong commercial and financial performance**
- **Created long-term value for patients and shareholders**

MERCK
INVENTING FOR LIFE

Significant progress over the past three years

Expanded and evolved our pipeline to drive **future innovations for patients**

Reached **more people globally** with our medicines and vaccines

Drove **innovation and productivity** through integration of data, digital and analytics

Strong Q2 sales[1] and earnings performance

Q2 Worldwide Sales

$16.1B

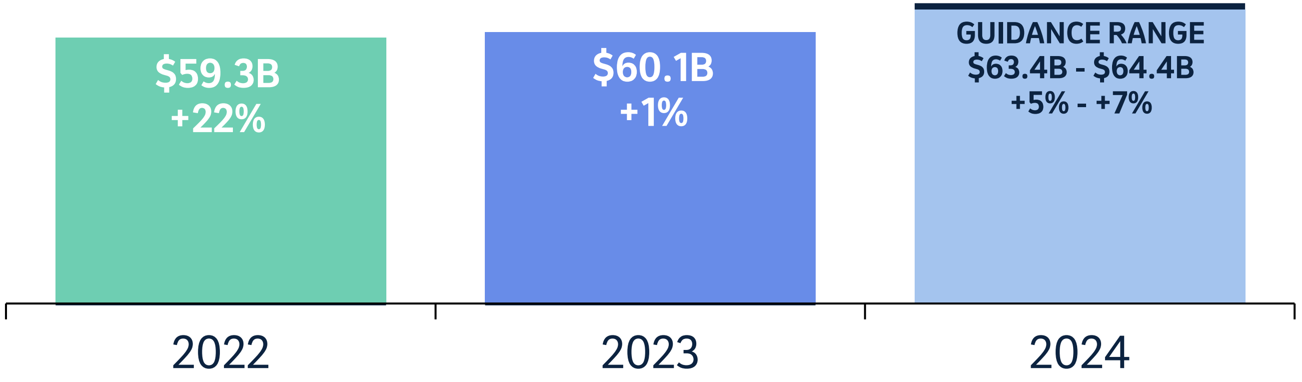
+7%

+11% ex-Exchange

Q2 Non-GAAP EPS[3]

$2.28

Full Year Sales



Full Year Non-GAAP EPS[2]

1. Results from continuing operations attributable to Merck & Co., Inc. 2. Merck does not exclude charges for certain upfront payments related to collaborations and licensing agreements, or charges related to pre-approval assets obtained in transactions accounted for as asset acquisitions from its non-GAAP results. Second quarter of 2023 includes $4.02 per share of such charges. Full year non-GAAP outlook for 2024 and full year reported results for 2023 and 2022 include $0.77, $6.21 and $0.22 per share of such charges, respectively. 3. GAAP EPS $2.14.

Advancing our broad pipeline

Cardiometabolic

WINREVAIR™
(sotatercept-csrk) for injection 45 mg, 60 mg

Successful initial launch following U.S. approval, and received positive CHMP opinion

Vaccines

CAPVAXIVE™
Pneumococcal 21-valent Conjugate Vaccine

Recent FDA approval and unanimous ACIP recommendation in certain adult populations

Oncology

Diverse Portfolio & Pipeline

Presented data for multiple novel candidates at ASCO

Business/Financial Results and Outlook

Caroline Litchfield
Chief Financial Officer

Strong Q2 worldwide sales growth

Merck

WORLDWIDE SALES[1,2]

$16.1B

+7% growth
+11% ex-exchange[3]

Human Health

$14.4B

+7% growth
+11% ex-exchange[3]

Animal Health

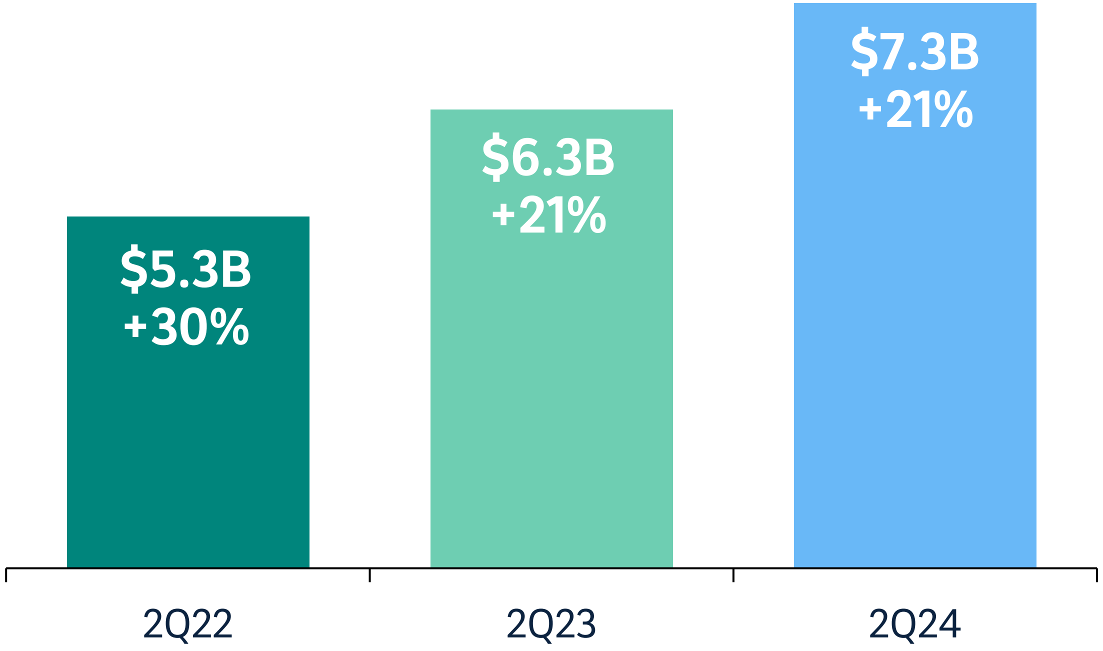
$1.5B

+2% growth
+6% ex-exchange[3]

1. Results attributable to Merck & Co., Inc. 2. Worldwide Sales includes Other Revenue.
3. ~2 percentage points of the negative impact of foreign exchange for Worldwide Sales and Human Health, and ~3 percentage points of the negative impact of foreign exchange for Animal Health, was due to devaluation of the Argentine peso, which was largely offset by inflation-related price increases, consistent with practice in that market.

Oncology: KEYTRUDA continues to drive excellent growth

- KEYTRUDA sales of $7.3B increased 21%[1] year-over-year, driven by uptake in earlier stage cancers and continued strong global demand from metastatic indications
  - In the U.S., increase reflects uptake from KN-671 and KN-091 in NSCLC, as well as uptake from KN-A39 in advanced urothelial cancer
  - Ex-U.S. growth reflects continued uptake in earlier stage cancers, including high-risk early-stage TNBC and adjuvant RCC, as well as demand from metastatic indications

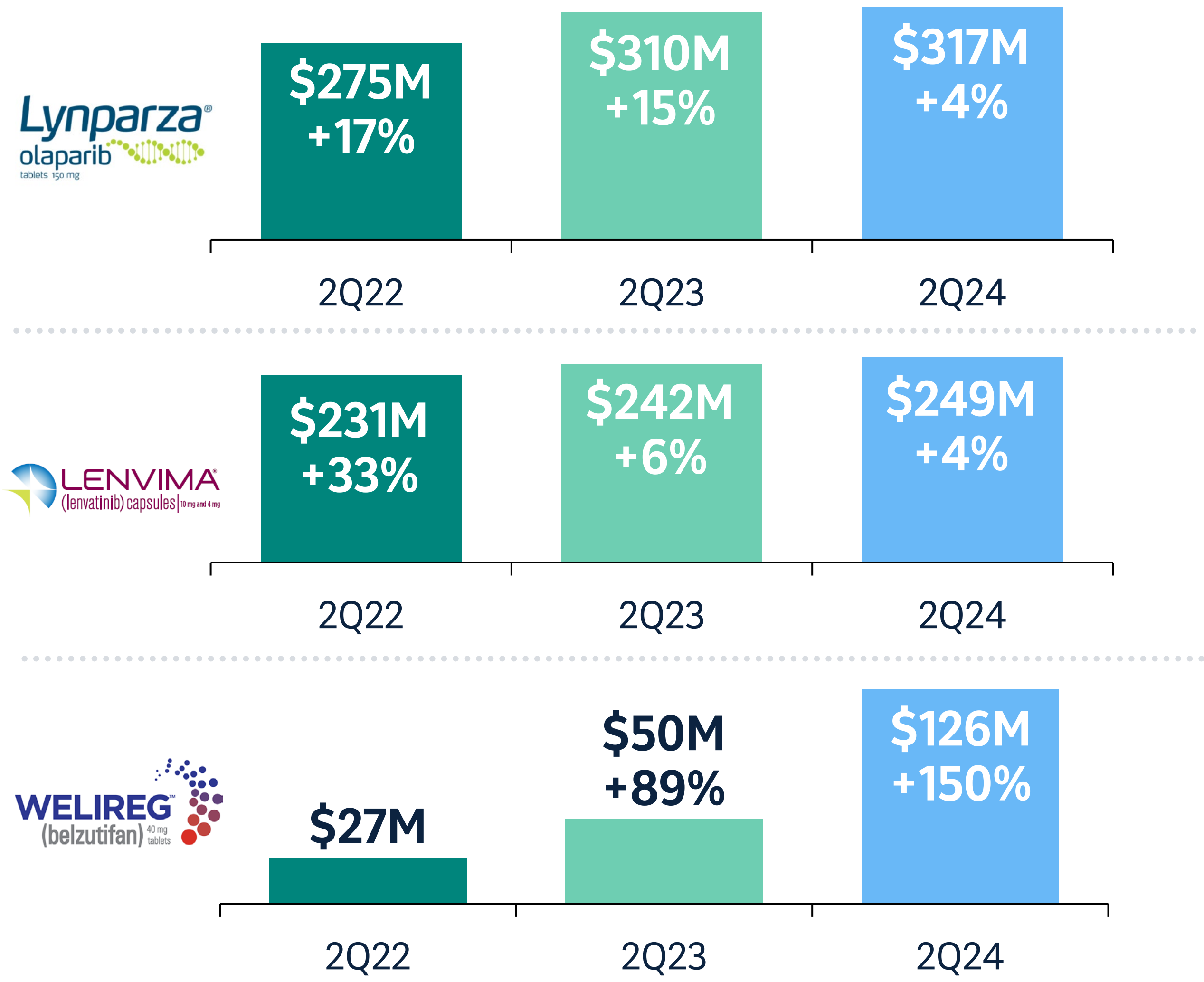


Growth rates exclude the impact of foreign exchange.

1. ~4 percentage point negative impact of foreign exchange was due to devaluation of Argentine peso, which was largely offset by inflation-related price increases, consistent with practice in that market.

Oncology: Strong performance across broad portfolio

- Lynparza[1] sales grew 4%, driven primarily by higher demand in the U.S. and certain international markets
- Lenvima[2] sales grew 4%, driven primarily by higher demand in the U.S.
- WELIREG sales more than doubled, driven by increased uptake in certain patients with previously treated advanced RCC

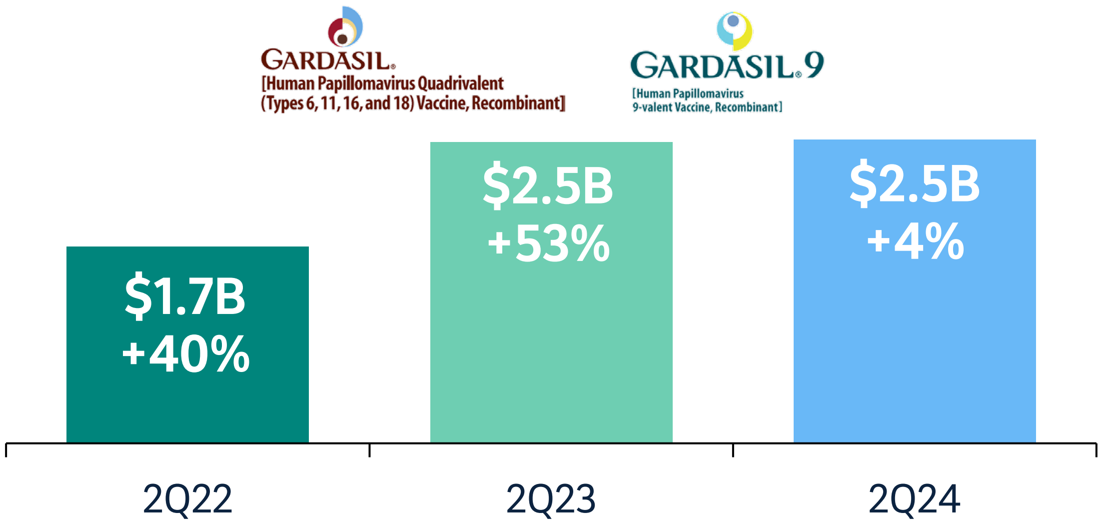


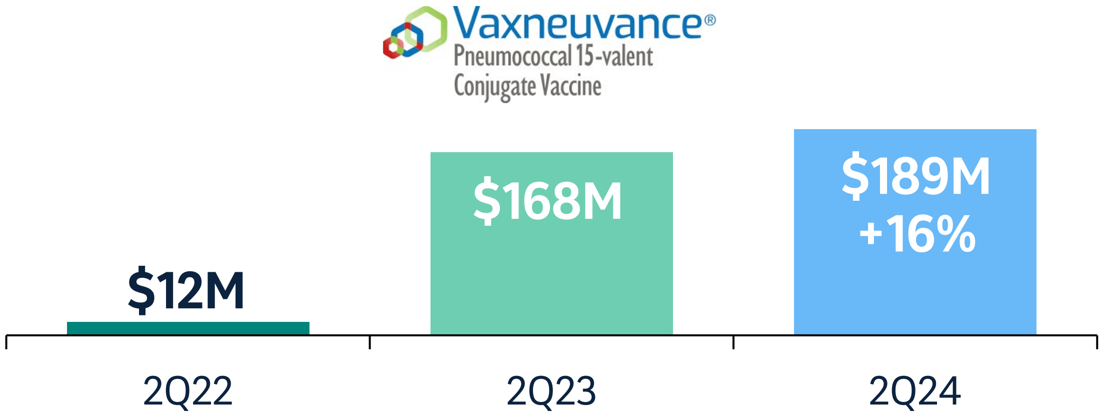
Growth rates exclude the impact of foreign exchange.
1. In collaboration with AstraZeneca 2. In collaboration with Eisai

Vaccines: Growth driven by GARDASIL and VAXNEUVANCE

- GARDASIL sales of $2.5B increased 4% year-over-year
  - In the U.S., sales benefitted from price, demand and favorable CDC purchasing patterns
  - Ex-U.S., sales were driven by higher demand across several markets, partially offset by timing of shipments in China

- VAXNEUVANCE grew to $189M, driven by ongoing launches in ex-U.S. markets





Growth rates exclude the impact of foreign exchange.

Cardiovascular: Successful initial launch of WINREVAIR

WINREVAIR™ (sotatercept-csrk) for injection 45 mg, 60 mg

Sales of $70M[1]

Patients

- **>2,000 patients prescribed** in the quarter
- Across all prescriptions, **~75-80% of patients receive commercial product**
- **>1,000 patients started treatment** in the quarter, largely reflecting **prescriptions written in April and May**
- Takes **~1 month** on average for patients to **start therapy**

Prescribers

- **>500 physicians** have written at least one prescription
- Many prescribers are looking to **gain experience with the product** as they **prioritize patients who are in greatest need** of additional therapy
- Most prescribers are from **large academic centers** or **larger private practices**

Payors

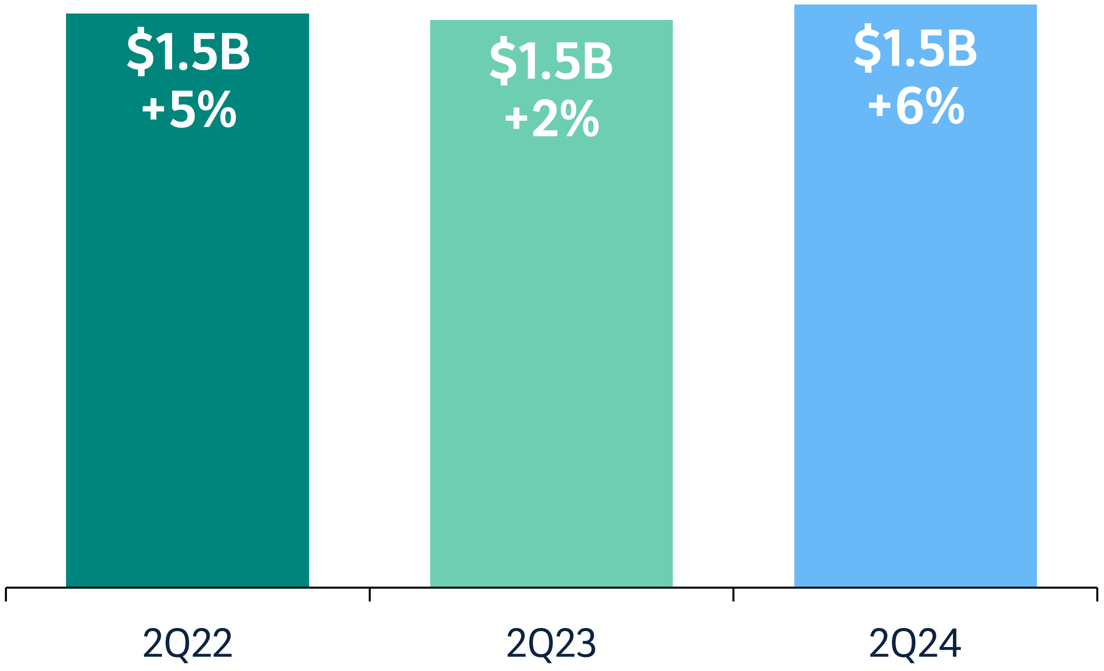
- **Many payors established coverage policies** consistent with the label and STELLAR criteria
- **Other payors** are in the process of **developing policies**

1. ~40% of sales were attributed to doses administered to patients, with the remainder due to distributors building inventory in support of increasing demand.

Animal Health: Growth driven by livestock

- Animal Health sales increased 6%[1] to $1.5B
  - Livestock sales grew 11%, driven by higher demand for poultry and ruminant products, as well as price
  - Companion Animal growth reflects price, partially offset by reduction in distributor inventory

MERCK Animal Health



Growth rates exclude the impact of foreign exchange.
1. ~3 percentage point negative impact of foreign exchange was due to devaluation of Argentine peso, which was largely offset by inflation-related price increases, consistent with practice in that market.

Q2 2024 non-GAAP financial results summary[1]

$ in billions, except EPS amounts

	Q2 2024	Q2 2023	Change	Change Ex-FX
Sales	$16.1	$15.0	+7%	+11%
Non-GAAP Gross Margin	80.9%	76.6%	+4.3pts	+4.3pts
Non-GAAP Operating Expenses	$6.2	$15.9	-61%	-61%
Non-GAAP Tax Rate	14.1%	-18.4%	+32.5pts	N/A
Non-GAAP Earnings / (Loss) Per Share[2,3]	$2.28	($2.06)	N/M	N/M

1. Merck is providing certain 2024 and 2023 non-GAAP information that excludes certain items because of the nature of these items and the impact they have on the analysis of underlying business performance and trends. Management believes that providing this information enhances investors' understanding of the company's results because management uses non-GAAP results to assess performance. Management uses non-GAAP measures internally for planning and forecasting purposes and to measure the performance of the company along with other metrics. In addition, annual employee compensation, including senior management's compensation, is derived in part using a non-GAAP pretax income metric. This information should be considered in addition to, but not as a substitute for or superior to, information prepared in accordance with GAAP. For a description of the non-GAAP adjustments, see Table 2a attached to the earnings release. 2. Q2 2023 includes a charge of $10.2 billion, or $4.02 negative EPS impact, for the acquisition of Prometheus Biosciences. 3. Q2 2024 GAAP EPS of $2.14.

Updated 2024 financial outlook

	Prior Guidance	Updated Guidance	Key Assumptions
Revenue	$63.1B to $64.3B	$63.4B to $64.4B	• Assumes ~3 percentage point FX headwind • Implies growth of 5% to 7%
Non-GAAP Gross Margin Rate	~81.0%	~81.0%	
Non-GAAP Operating Expenses[1]	$25.2B to $26.1B	$26.8B to $27.6B	• Includes incremental $1.5B of charges related to EyeBio acquisition and advancing assets, as well as investments in progressing innovative pipeline
Other (Income) / Expense	~$250M of expense	~$350M of expense	• Includes financing costs for the acquisitions of EyeBio and Elanco's aqua business
Tax Rate	~14.5% to 15.5%	~15.5% to 16.5%	• Includes unfavorable impact from the non-tax deductible upfront charge related to acquisition of EyeBio
Shares Outstanding	~2.55B	~2.54B	• Assumes modest share repurchase
Non-GAAP EPS	$8.53 to $8.65	$7.94 to $8.04	• Includes one-time ~$0.51 charge related to EyeBio acquisition, as well as ~$0.09 to finance EyeBio and Elanco aqua business acquisitions and advance the acquired assets • Assumes more than $0.30 FX headwind in 2024

1. Guidance does not assume any additional significant potential business development transactions.

Key modeling considerations

CAPVAXIVE

- Received FDA approval and ACIP recommendation
- Commercial uptake will be enabled by milestones such as MMWR publication, which typically lags ACIP recommendation by a few months, as well as payor coverage and customer contracting

GARDASIL

- Benefitted from extremely strong demand of GARDASIL in China over the past few years
- In 2Q there was a significant step down in shipments from Zhifei to points of vaccination compared to prior quarters, resulting in higher inventory levels at Zhifei
- If shipments from Zhifei do not increase, it is likely that we will ship less than our full year 2024 contracted doses[1] by the end of this year
- Opportunity in China remains very attractive as there are >120M females in the addressable population living in Tier 1-5 cities have not yet received an HPV vaccination

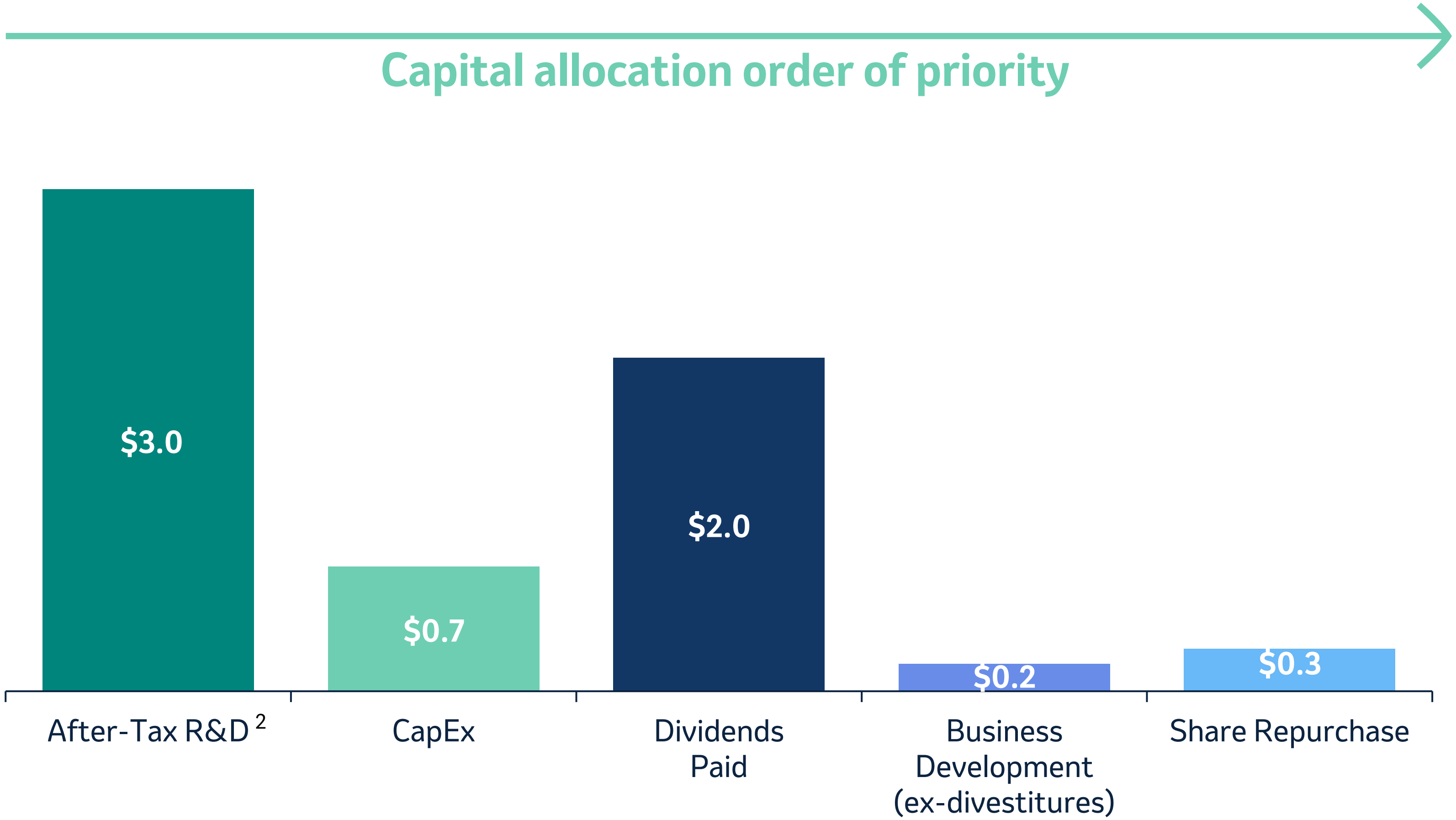
WINREVAIR

- Received CHMP opinion and anticipate EMA approval in the coming months
- Reimbursement should occur in 2025 in most major markets, but expect Germany will receive reimbursement and launch this year

1. Contracted doses in 2024 are less than the doses that were shipped in 2023

Remain committed to balanced capital allocation strategy

Q2 Spend ($ in billions)



Continue to invest in the **pipeline** and **business** while augmenting the pipeline with value-enhancing **business development**

1. Reflects quarter spend
2. Reflects R&D excluding Business Development

Research Update

Dr. Dean Li
President, Merck Research Laboratories

Important progress across our immunization programs

Pneumococcal

- Received **FDA approval** for **CAPVAXIVE**, a 21-valent pneumococcal conjugate vaccine for adults, as well as **unanimous ACIP recommendation** for certain adult populations
- **First approved pneumococcal conjugate vaccine** specifically designed to **cover serotypes responsible for ~85% of incidence of invasive pneumococcal disease** in individuals 65 years and older[1]
- Currently **under review by EMA**

CAPVAXIVE™
Pneumococcal 21-valent
Conjugate Vaccine

RSV

- Announced **positive topline results** from **Phase 2b/3 trial evaluating clesrovimab**, an investigational RSV preventative monoclonal antibody for infants
- **Study met primary efficacy and safety endpoints**, as well as **secondary endpoint** regarding RSV-associated hospitalization
- Potential to provide a **single, fixed-dose option** to help **protect healthy and high-risk infants** from birth through their first full RSV season

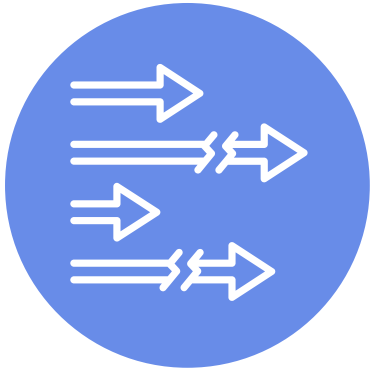
1. According to CDC data from 2018-2021

Executing upon our oncology development strategy

Immuno-oncology

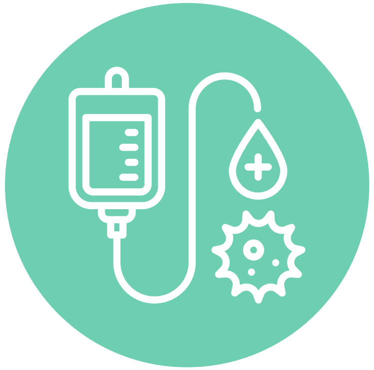
Boost anti-tumor immune responses

Precision Molecular Targeting



Impact pathways that can drive cancer growth

Tissue Targeting



Increase cancer cell sensitivity with ADCs and immune-engagers

Continuing to harness the power of immuno-oncology

KEYTRUDA®
(pembrolizumab) Injection 100 mg

Earlier Stage:	Total:
9 approvals[1] in distinct indications	**40** approvals[1] in distinct indications
4 studies with OS benefit	**25** studies with OS benefit

Recent KEYTRUDA Updates

- **KEYNOTE-868:** FDA approved KEYTRUDA in combination with chemotherapy for treatment of primary advanced or recurrent endometrial cancer regardless of mismatch repair status
- **KEYNOTE-522:** Announced study demonstrated OS benefit in patients with high-risk early-stage TNBC
- **KEYNOTE-811:** Announced study met OS dual primary endpoint for treatment of 1L HER2-positive advanced gastric or GEJ adenocarcinoma
- **KEYNOTE-483:** FDA granted priority review for KEYTRUDA in combination with chemotherapy for 1L treatment of unresectable advanced or metastatic malignant pleural mesothelioma (PDUFA September 25, 2024)

1. Represents approvals in the U.S.

Updates across precision and tissue targeting

Precision Molecular Targeting

- Exercised option to advance **opevesostat**[1], an oral nonsteroidal **CYP11A1** inhibitor
- **Two ongoing Phase 3** trials, **OMAHA-1 and OMAHA-2,** evaluating **opevesostat**[1] in combination with hormone replacement therapy for treatment of certain patients with **metastatic prostate cancer**

Omaha
ONCOLOGY CLINICAL TRIALS

Tissue Targeting

- Received **positive CHMP opinion** for **KEYTRUDA in combination with Padcev** for 1L treatment of patients with **unresectable or metastatic urothelial carcinoma** based on KN-A39/EV-302[2]
- FDA issued a Complete Response Letter for BLA for **patritumab deruxtecan**[3] for treatment of certain adult patients with **locally advanced or metastatic EGFR-mutated NSCLC** previously treated with two or more systemic therapies

1. In collaboration with Orion 2. Trial conducted in collaboration with Seagen (now Pfizer) and Astellas 3. In collaboration with Daiichi Sankyo

Important developments across our diverse pipeline

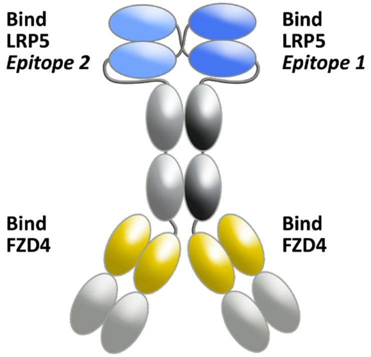
Cardiometabolic

- Received **positive CHMP opinion** for **WINREVAIR** for treatment of PAH
- Anticipate **European Commission decision** on marketing authorization application in **3Q 2024**

Ophthalmology

- Completed **acquisition of EyeBio**
- **Restoret™/MK-3000** is an investigational, potentially first-in-class tetravalent tri-specific **Wnt antibody** for treatment of **diabetic macular edema and neovascular age-related macular degeneration**

Restoret™/MK-3000[1]:



Q&A

Rob Davis
Chairman & Chief Executive Officer

Caroline Litchfield
Chief Financial Officer

Dr. Dean Li
President, Merck Research Laboratories

Peter Dannenbaum
Senior Vice President, Investor Relations

Appendix

Q2 2024 GAAP financial results summary

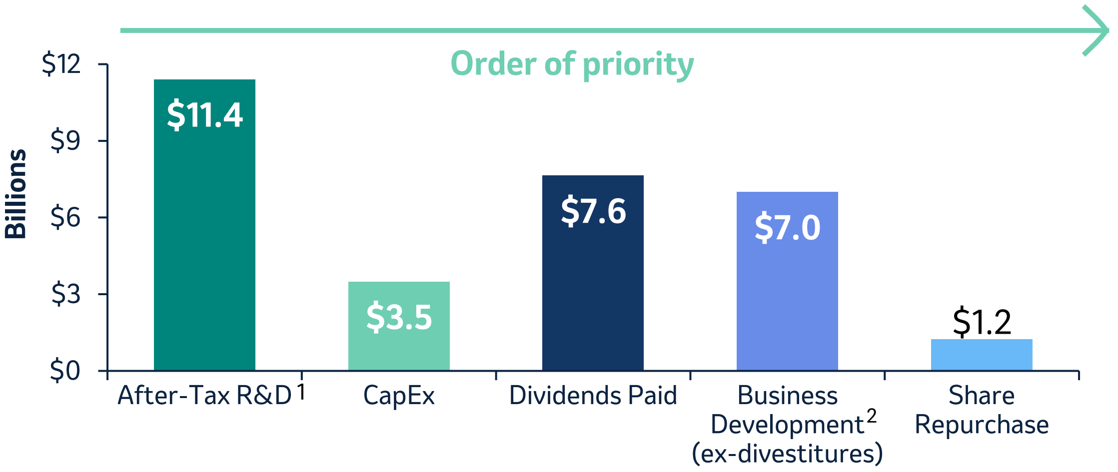
$ in billions, EPS amounts

	Q2 2024	Q2 2023	Change	Change Ex-FX
Sales	$16.1	$15.0	+7%	+11%
Operating Expenses[1]	$6.2	$16.0	-61%	-60%
Tax Rate	9.1%	-11.9%	-21.0pts	N/A
GAAP Earnings / (Loss) per Share[1]	$2.14	($2.35)	N/M	N/M

1. 2Q23 GAAP results include $10.2 billion charge or $4.02 per share for the acquisition of Prometheus Biosciences

Capital allocation: Trailing twelve months

Over the past 12 months



Well-positioned balance sheet with capacity to fund **additional value-enhancing business development** opportunities

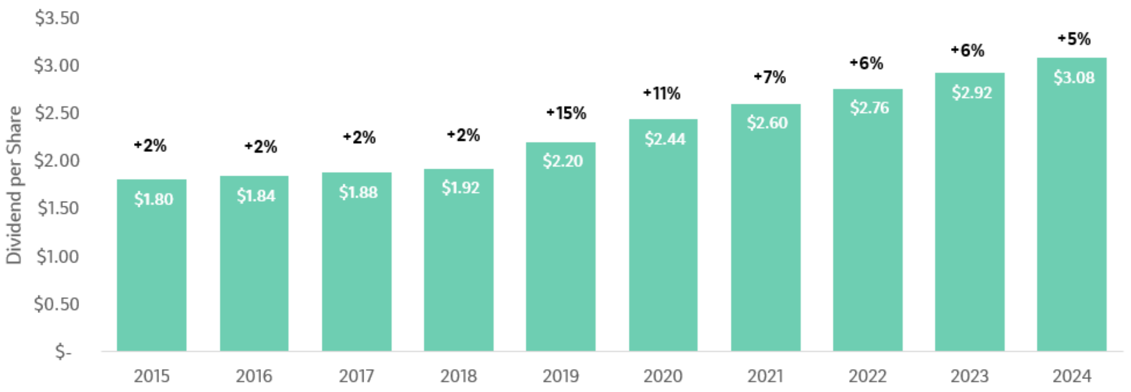
Capital investments

2024 to 2028

~$19B[3]

Over 5 years, including expanding manufacturing capacity for Oncology, Vaccines, and Animal Health. Includes >$11B in the U.S.

Commitment to the dividend



1. Reflects R&D excluding Business Development
2. Includes BD payments reflected in operating cash flow
3. Updated October 29, 2024: Previous values corrected by the company

Driving value for patients and shareholders by progressing our pipeline

Key regulatory milestones since the last earnings call:

- **In the U.S.:**
  - FDA accepted for priority review the sBLA for KEYTRUDA in combination with chemotherapy as first-line treatment of patients with unresectable advanced or metastatic malignant pleural mesothelioma based on IND.227/KN-483
  - FDA approved KEYTRUDA in combination with chemotherapy, followed by KEYTRUDA as a single agent, for the treatment of adult patients with primary advanced or recurrent endometrial carcinoma based on NRG-GY018/KN-868
  - FDA approved CAPVAXIVE for the prevention of invasive pneumococcal disease and pneumococcal pneumonia in adults, and CDC's ACIP unanimously voted to recommend CAPVAXIVE as an option for certain adults
- **In the EU:**
  - Received positive CHMP opinion for KEYTRUDA in combination with Padcev for the first-line treatment of adult patients with unresectable or metastatic urothelial carcinoma based on KN-A39/EV-302[1]
  - Received positive CHMP opinion for WINREVAIR, in combination with other PAH therapies, for the treatment of PAH in adult patients with WHO FC II to III

Key data & clinical advancements since the last earnings call:

- Presented data across broad oncology portfolio at ASCO, including for KEYTRUDA (KN-966, KN-859, KN-A39, KN-671, KN-224, KN-629), mRNA-4157/V940[2] (INT), vibostolimab (anti-TIGIT), sacituzumab tirumotecan[3] (TROP2 ADC), and patritumab deruxtecan[4] (HER3 ADC)
- Announced Phase 2b/3 trial evaluating clesrovimab (MK-1654) met its primary safety and efficacy endpoints, including reducing medically attended lower respiratory infections (MALRI) caused by RSV through Day 150
- Announced Phase 3 KEYNOTE-811 and KEYNOTE-522 trials for KEYTRUDA met overall survival endpoints
- Advanced our broad pipeline, initiating Phase 3 trials across multiple candidates including sacituzumab tirumotecan[3] (MK-2870, TROP2 ADC), enlicitide decanoate (MK-0616, PCSK9 inhibitor), and tulisokibart (MK-7240, TL1A inhibitor)

1. Trial conducted in collaboration with Seagen (now Pfizer) and Astellas 2. In collaboration with Moderna
3. In collaboration with Kelun-Biotech 4. In collaboration with Daiichi Sankyo

Broad and innovative pipeline to address significant unmet medical needs

Phase 2

Oncology

MK-1022 (patritumab deruxtecan)[6]
Gastric
HNSCC
Melanoma

MK-1308 (quavonlimab)
NSCLC

MK-1308A (quavonlimab +pembrolizumab)
CRC

MK-2140 (zilovertamab vedotin)
Hematological Malignancies

MK-2400 (ifinatamab deruxtecan)[6]
Bladder
CRC
Endometrial
HNSCC

MK-2870 (sacituzumab tirumotecan)[7]
Biliary
CRC
Neoplasm Malignant
Pancreatic

KEYTRUDA (MK-3475)
Advanced Solid Tumors
Prostate

MK-3475A (pembrolizumab + hyaluronidase)
cSCC

MK-4280 (favezelimab)
NSCLC

MK-4280A (favezelimab+pembrolizumab)
Bladder
cSCC
Endometrial
Esophageal
Melanoma
RCC

MK-5890 (boserolimab)
Neoplasm Malignant

MK-5909 (raludotatug deruxtecan)[6]
Ovarian

LYNPARZA (MK-7339)[8]
Advanced Solid Tumors

LENVIMA (MK-7902)[9]
HNSCC

WELIREG (MK-6482)
Endometrial
Esophageal
HCC
Prostate
Rare Cancers

MK-7684A (vibostolimab +pembrolizumab)
Bladder
CRC
Endometrial
Melanoma
Ovarian
Prostate
RCC

V940[10]
Bladder
cSCC
RCC

Vaccines

V181
Dengue Virus

Infectious disease

MK-8527
HIV-1 prevention

MK-8591B (islatravir+MK-8507)[1]
HIV-1 infection

MK-8591D (islatravir+lenacapavir)[2,11]
HIV-1 infection

Cardiometabolic

MK-2060
Thrombosis

MK-5475
PH-COPD

MK-6024 (efinopegdutide)
NASH

WINREVAIR (MK-7962)
Pulmonary Hypertension due to Left Heart Disease

Neuroscience

MK-8189[4]
Schizophrenia

Immunology

MK-6194
Vitiligo

Ophthalmology

Restoret™/MK-3000
Diabetic macular edema

Phase 3

Oncology

MK-1022 (patritumab deruxtecan)[6]
NSCLC (EU)

MK-1026 (nemtabrutinib)
Hematological Malignancies

MK-1084
NSCLC

MK-1308A (quavonlimab +pembrolizumab)
RCC

MK-2400 (ifinatamab deruxtecan)[6]
SCLC

MK-2870 (sacituzumab tirumotecan)[7]
Breast
Cervical
Gastric
Endometrial
NSCLC

MK-3543 (bomedemstat)
Myeloproliferative Disorders

KEYTRUDA (MK-3475)
CSCC (EU)
Hepatocellular (EU)
Ovarian
SCLC

MK-3475A (pembrolizumab +hyaluronidase)
NSCLC

MK-4280A (favezelimab +pembrolizumab)
CRC
Hematological Malignancies

MK-5684 (opevesostat)[13]
Prostate

LYNPARZA (MK-7339)[8]
NSCLC
SCLC

LENVIMA (MK-7902)[9]
Esophageal
Gastric

MK-7684A (vibostolimab +pembrolizumab)
NSCLC
SCLC

V940[10]
Melanoma
NSCLC

Immunization

MK-1654 (clesrovimab)
Respiratory Syncytial Virus (RSV)

Infectious disease

MK-8591A (doravirine+islatravir)[2]
HIV-1 Infection

LAGEVRIO (MK-4482)[3,12]
COVID-19 antiviral

Immunology

MK-7240 (tulisokibart)
Ulcerative Colitis

Cardiometabolic

MK-0616 (enlicitide decanoate)
Hypercholesterolemia

Under regulatory review

Oncology

KEYTRUDA (MK-3475)
Resectable NSCLC (JPN)
1L Urothelial (EU, JPN)
Endometrial Carcinoma (EU, JPN)
Cervical (EU, JPN)
Mesothelioma (US, EU, JPN)

MK-1022 (patritumab deruxtecan)[6, 14]
NSCLC (US)

WELIREG (MK-6482)
Certain VHL tumors (EU)
Advanced RCC (EU, JPN)

General Medicine

Gefapixant (MK-7264)[5]
Cough (US)

Cardiometabolic

WINREVAIR (MK-7962)
Pulmonary Arterial Hypertension (EU)

Vaccines

CAPVAXIVE (V116)
Pneumococcal conjugate vaccine, adult (EU)

[1] On FDA clinical hold [2] On FDA partial clinical hold for higher doses than those used in current clinical trials
[3] Available in the US under EUA [4] Development is co-funded by Royalty Pharma [5] FDA issued CRL in December 2023
[6] In collaboration with Daiichi Sankyo [7] In collaboration with Kelun-Biotech [8] In collaboration with AstraZeneca [9] In collaboration with Eisai
[10] In collaboration with Moderna [11] In collaboration with Gilead [12] Developed under an agreement with Ridgeback Bio [13] In collaboration with Orion
[14] FDA issued CRL in June 2024

As of July 30, 2024